
CSCI 441/541 Neural Network and Deep Learning

BASIC INFORMATION

Course Title: Neural Network and Deep Learning
Course designator: CSCI 441/541
Section: 54 (Online)
Number of credits: 03
Term and year: Fall 2026
Department: CIDS
Instructor's name: Maninder Singh
Office location: CH 366W
Support hours:

Time	Monday	Tuesday	Wednesday	Thursday
9 AM – 9:50 AM	Office hour via (Zoom)	Office hour via (Zoom)	Office hour via (Zoom)	Office hour via appointment (Zoom)
10 AM – 10:50 AM	Office hour via (Zoom)	Office hour via (Zoom)	Office hour via (Zoom)	Office hours via email appointment (Zoom)
11AM – 11:50 AM	Office hour via (Zoom)		Office hour via (Zoom)	

Join Zoom Meeting for appointments

<https://minnstate.zoom.us/j/98263954472>

Meeting ID: 982 6395 4472

Password: 988202

Contact Information:

Email: msingh@stcloudstate.edu (preferred)

You should expect to receive the reply of your query (via email) approximately around 24-48 hours except over weekend and non-meeting days. Your query should have the following format of subject line in your email:

(Course Name) - (Your Name)-Short_Query_Subject_Line

Course Times and Location: **Asynchronous:** The recorded lectures will be posted in D2L.

Days of Week: NA

Location: NA

BRIEF DESCRIPTION OF THE COURSE

Concepts and techniques in deep learning, exploring both theoretical foundations and practical applications. Neural network models and deep learning methods for complex real-world problems. Emphasis on critical thinking, innovation, and hands-on experience with state-of-the-art tools and frameworks.

COURSE LEARNING OUTCOMES

Upon the successful completion of this course, students will be able to:

1. Design, train, and apply different forms of neural networks for data analysis.
2. Implement and evaluate regularization techniques, such as dropout, to improve model generalization and prevent overfitting.
3. Design and apply convolutional neural networks (CNNs) to image classification, object detection, and other computer vision applications.
4. Apply long short-term memory (LSTM) networks to speech recognition, machine translation, and time series analysis.
5. Apply the architecture of transformers and large language models to cutting-edge tasks such as natural language understanding, machine translation, and text generation.

For the graduate section (CSCI 541 in addition to the above five outcomes)

6. Design and evaluate an experimental approach in deep learning to investigate a selected research problem.

TEXT AND MATERIALS

As such, there is no specific book targeted as a “main textbook” for this course, as I will refer to multiple sources, including online content and textbooks. I will recommend where you can find the content, but you are also welcome to refer to any textbook or content on Deep Learning that covers our topics of

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discussion. My lecture notes (as applicable) will be available to you via D2L. The following are a few renowned textbooks I will refer to for this course.

1. Deep Learning
 - a. Ian Goodfellow, Yoshua Bengio, Aaron Courville
 - b. MIT Press
 - c. ISBN-10: 0262035618
 - d. ISBN-13: 978-0262035613
2. Neural Networks and Deep Learning (Second Edition)
 - a. Charu C. Aggarwal
 - b. Springer Nature
 - c. ISBN: 978-3-031-29641-3
 - d. ISBN (ebook): 978-3-031-29642-0
3. Hands-On Machine Learning with Scikit-Learn, Keras, and Tensor Flow (3rd edition onwards)
 - a. Geron, Aurelien
 - b. INGRAM publisher
 - c. ISBN 13: 9781098125974

ASSIGNMENTS

There will be five (5) home assignments on the different topics covered in the class. More information about the assignments is discussed as follows:

Home assignments:

These assignments will require more time and effort. They will be available in D2L at the beginning of each week. The assignments should be submitted via D2L (unless other submission instructions are provided). All the assignments will require an individual submission. It is up to the pupil to test their work thoroughly to ensure it is correct and functions properly.

D2L Quizzes:

There will be a minimum of 4 quizzes in D2L, and instructions will be provided in advance.

Project:

This course will include one project. The project details will be shared with you within the first four weeks of the course.

Due date and late penalties:

Each assignment will have a due date and time, listed along with the assignment. If your assignment is submitted after the due time, it will be considered late and incur a penalty. Weekends and holidays are counted as late days. No assignment will be accepted more than one day past the due date. Late submission will have a 30% late penalty. Please note that an assignment submitted even 1 minute past the due date is automatically labeled as late and will incur a 30% penalty (no excuses; avoid last-minute submissions). Wrong submissions and submissions into the wrong dropbox are considered “no

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submission” and will result in a zero. So, make sure you check what you turned in before considering the assignment submission complete.

TENTATIVE ASSIGNMENT SCHEDULE

Check D2L for assignment schedule.

TENTATIVE QUIZZES SCHEDULE

Check D2L for the exact quiz schedule.

EXAMS

There will be two exams for this course: the mid-term and the final, and they will cover the topics covered in class. The exams will be available via D2L. The exam will typically consist of questions similar to those found on the preceding quizzes or presented to you during an exam review and will have a similar structure. All exams will be available online in D2L. Additional instructions will be posted in D2L.

GRADING

Grading will be based on the following scale:

A+	97% - 100%
A	93% - 96.99%
A-	90% - 92.99%
B+	87% - 89.99%
B	83% - 86.99%
B-	80% - 82.99%
C+	77% - 79.99%
C	73% - 76.99%
C-	70% - 72.99%
D+	67% - 69.99%
D	63% - 66.99%
D-	60% - 62.99%
F	59.99% or below

Your grade consists of the following graded items:

1. Assignments	20%
2. D2L Quizzes	10%
3. Project	20%
4. Exam-1 (Comprehensive)	20%
5. Exam-2 (Comprehensive)	30%

AI USE POLICY

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Supported tools, such as Copilot and Adobe AI tools, may be used in this course for limited academic tasks, such as idea generation, outlining, or early-stage revisions. However, all submitted work must reflect your own understanding, voice, and academic reasoning.

AI tools should serve as a support, not a replacement, for your thinking. Submitting work that is primarily AI-generated without acknowledgment is a breach of our Academic Integrity Policy.

If you use AI tools as part of your process, you must be transparent about how they were used (e.g., “Used Copilot to draft bullet points; expanded and revised with personal analysis”). When in doubt, ask for clarification before submitting your work.

Please consult the Academic Integrity Policy for more information, and let’s maintain an open dialogue about ethical and responsible use of technology.

SCHOLASTIC DISHONESTY

You are expected to do your own academic work and cite sources as necessary. Failing to do so is scholastic dishonesty. Scholastic dishonesty means plagiarizing; cheating on assignments or examinations; engaging in unauthorized collaboration on academic work; taking, acquiring, or using test materials without faculty permission; submitting false or incomplete records of academic achievement; acting alone or in cooperation with another to falsify records or to obtain dishonestly grades, honors, awards, or professional endorsement; altering, forging, or misusing a University academic record; or fabricating or falsifying data, research procedures, or data analysis.

If it is determined that student has cheated, he or she may be given an "F" for the course, and may face additional sanctions from the University.

The Office for Student Conduct and Academic Integrity has compiled a useful list of Frequently Asked Questions pertaining to scholastic dishonesty:

If you have additional questions, please clarify with your instructor for the course. Your instructor can respond to your specific questions regarding what would constitute scholastic dishonesty in the context of a particular class-e.g., whether collaboration on assignments is permitted, requirements and methods for citing sources, if electronic aids are permitted or prohibited during an exam.

STUDENT ACCESSIBILITY SERVICES

An affirmative action, equal opportunity employer and educator. St. Cloud State University is committed to a policy of nondiscrimination in employment and education opportunity and works to provide reasonable accommodations for all persons with disabilities. Accommodations are provided on an individualized, as-needed basis, determined through appropriate documentation of need. Please contact Student Accessibility Services sas@stcloudstate.edu or 320-308-4080, office Centennial Hall CH-202 to meet and discuss reasonable and appropriate accommodations for your plan. The accommodations authorized in your plan should be discussed with your instructor. All discussions will remain confidential. This syllabus is available in alternate formats upon request.

<https://www.stcloudstate.edu/sas/student-resources.aspx>

Additional Policies:

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- All assignments will be announced in class and posted on D2L. If you miss class for any reason, it is your responsibility to find out what assignments you missed.
- All electronic devices brought to class must have the sound turned off. If disruption of the class by devices such as cell phones or laptop computers occurs frequently, action will be taken to restore a productive learning environment.
- Headphones or other personal listening devices are not allowed during quizzes or exams.
- If you dispute the grading of any material, you have two weeks from the date the grade is recorded to request a change in the grade. After this time, no alterations will be considered, unless a grading error is discovered.
- More information about husky compact can be found at the following link:

Our Husky Compact: <https://www.stcloudstate.edu/ourhuskycompact/>

UPDATE HISTORY

Aug 05: Syllabus Creation